BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
July 2025 CSE 210 (Computer Architecture Sessional)

1 Introduction

As part of this assignment, you are required to submit both a software simulation and a hardware implementation of a simplified Arithmetic Logic Unit (ALU). The functional design specification for each group within each section are provided in *Appendix A*. This is a group assignment, and all group members are expected to participate equally.

The submission deadline for the software simulation is uniform across all sections and is set for January 3, 2026 (Saturday) at 11:59 PM. On the other hand, The hardware implementation is due on the next sessional day, which differs by section (refer to the *Deadline* section for details). Additionally, each group must prepare a report and submit it on the evaluation day.

2 Specification for 4-bit ALU Simulation

- The functional design specification for all groups can be found in Appendix A. First, read the specification of your group carefully. Please review the specification assigned to your group thoroughly before proceeding with the requirements detailed in this section.
- Design the ALU as efficiently as possible, minimizing the number of ICs used, in accordance with the assigned specification.
- In addition, you must also implement the following status flags:
  - Carry (C)
  - Sign (S)
  - Overflow (V)
  - Zero (Z)
- The flags should be updated according to conventional assembly language rules. However, for the purposes of this assignment, certain exceptions have been introduced to simplify flag handling for logical operations. Remember that this flexibility is to make your assignment easier, although it breaks some of the rules of the assembly programming language.

  1. For NOT Operation:
  - After the NOT operation, which makes the result to 0000, if Z becomes 0 from 1, it will not be accepted. However, if Z becomes 1 or if Z value remains unchanged, both will be accepted.
  - After the NOT operation, if S remains unchanged or it reflects the highest order bit of the result, both will be accepted. But if S bit changed and it changed to a wrong value, it will not be accepted.

- To make your life easier, we shall not check the C and V bits after NOT operation, i.e., you can consider these as don't care.

2. For AND/OR/XOR Operation:

- C and V should be cleared (0) after the operation.
- S and Z should be changed according to the output.

- Any 2-input SSI (AND, OR, NOT, XOR etc.) and MSI (MUX, Decoder, Adder etc.) chip can be used.
- Emphasis should be given on efficiency of design and minimization of ICs used.
- For simulation, you can use any simulation software.
- Your software design must be in IC-level.
- **Software Simulation Submission:** A submission link will be opened on Moodle for submitting your ALU simulation. Make a folder containing all your simulation project files, zip it and submit it following the naming format. If you submit only a single project file, then just submit the file following the naming format. The naming format should be your section name followed by your group id (e.g., B1_Group_5). Please ensure a single submission from each group (only a single member of the group should submit).
- **Report Preparation Guideline:** Your report must contain:
  - Introduction
  - Problem Specification with assigned instructions
  - Detailed design steps with k-maps (if applicable)
  - Truth Table
  - Block Diagram
  - Complete Circuit Diagram
  - ICs used with count as a chart
  - Simulator used along with the version number
  - Discussion

  The report must be handwritten.
- **Software Simulation, Hardware Implementation Evaluation and Report Submission:** Both software and hardware components will be evaluated during the regular laboratory session for each section (See the *Deadline* section for the specific time and date). Each group must bring at least one laptop to demonstrate the software simulation. Also, you need to submit your report at the beginning of the sessional.
- For hardware implementation, you can lend the required instruments from the laboratory from now on (on the condition of returning these immediately after the evaluations and without any alteration), or you can use your own instruments.
- Any type of plagiarism will be punished.
- All the specified date and time is according to the Bangladesh Time. Late submission is not allowed.

3 Deadline

Software Simulation Submission Deadline: For all sections: January 3, 2026 (Saturday) at 11:59 PM BDT.

Software Simulation and Hardware Evaluation Timeline

- A1: January 5, 2026 (2:30 PM)
- A2: January 12, 2026 (2:30 PM)
- B1: January 7, 2026 (2:30 PM)
- B2: January 14, 2026 (2:30 PM)
- C1: January 4, 2026 (2:30 PM)
- C2: January 11, 2026 (2:30 PM)

4 Where to Ask?

For any queries, you may consult your instructor during theory or sessional classes. You may also send an email to `tahmid@teacher.cse.buet.ac.bd`.

5 Version

This section documents the versioning of the assignment description. The document begins at Version 0. In the event of major corrections or revisions, the version number will be incremented, and all changes will be listed here. Please monitor the version number on Moodle to determine whether an update applies to your group. Minor edits, such as grammatical fixes, will not trigger a version update.

5.1 Version 0

This is the initial version of the problem description pdf.

Appendix A

Functional Design Specifications for All Groups

Section A1

Control Signals			Functions for					
cs2	**cs1**	**cs0**	**Group 1**	**Group 2**	**Group 3**	**Group 4**	**Group 5**	**Group 6**
0	0	0	Decrement A	Sub with borrow	Add	Add with carry	Add with carry	Decrement A
0	0	1	Add with carry	Sub	Increment A	Add	Sub	Add
1	0	0	Sub with borrow	Increment A	Transfer A	Decrement A	Decrement A	Increment A
1	0	1	Increment A	Add	Sub with borrow	Increment A	Transfer A	Add with carry
X	1	0	XOR	OR	OR	AND	AND	XOR
X	1	1	NOT	AND	XOR	OR	XOR	OR

Section A2

Control Signals			Functions for					
cs2	**cs1**	**cs0**	**Group 1**	**Group 2**	**Group 3**	**Group 4**	**Group 5**	**Group 6**
0	0	0	Sub	Increment A	Add with carry	Sub	Add	Transfer A
0	1	1	Decrement A	Decrement A	Sub with borrow	Add with carry	Increment A	Sub with borrow
1	0	0	Increment A	Transfer A	Sub	Increment A	Add with carry	Sub
1	1	1	Add with carry	Add with carry	Decrement A	Decrement A	Sub with borrow	Increment A
X	0	1	AND	XOR	OR	AND	NOT	AND
X	1	0	OR	NOT	XOR	XOR	AND	NOT

Section B1

Control Signals			Functions for					
cs2	**cs1**	**cs0**	**Group 1**	**Group 2**	**Group 3**	**Group 4**	**Group 5**	**Group 6**
0	0	1	Transfer A	Add with carry	Add	Decrement A	Increment A	Transfer A
0	1	0	Add with carry	Transfer A	Sub	Increment A	Sub with borrow	Increment A
1	0	1	Add	Add	Add with carry	Sub	Sub	Add with carry
1	1	0	Decrement A	Sub	Increment A	Transfer A	Transfer A	Add
X	0	0	XOR	NOT	XOR	XOR	NOT	AND
X	1	1	NOT	XOR	OR	OR	OR	OR

Section B2

Control Signals			Functions for					
cs2	**cs1**	**cs0**	**Group 1**	**Group 2**	**Group 3**	**Group 4**	**Group 5**	**Group 6**
0	0	1	Sub	Transfer A	Transfer A	Sub	Decrement A	Increment A
0	1	1	Add with carry	Decrement A	Increment A	Transfer A	Sub with borrow	Sub with borrow
1	0	1	Decrement A	Sub with borrow	Decrement A	Decrement A	Add with carry	Add with carry
1	1	1	Transfer A	Sub	Add with carry	Add with carry	Add	Sub
X	0	0	NOT	XOR	NOT	AND	XOR	NOT
X	1	0	AND	OR	AND	OR	AND	OR

Section C1

Control Signals			Functions for					
cs2	**cs1**	**cs0**	**Group 1**	**Group 2**	**Group 3**	**Group 4**	**Group 5**	**Group 6**
0	0	0	Add	Transfer A	Add with carry	Add	Add	Add
0	1	0	Add with carry	Sub with borrow	Transfer A	Transfer A	Transfer A	Add with carry
1	0	0	Sub with borrow	Add	Add	Add with carry	Decrement A	Decrement A
1	1	0	Transfer A	Decrement A	Sub with borrow	Sub with borrow	Sub with borrow	Transfer A
X	0	1	XOR	AND	NOT	NOT	XOR	NOT
X	1	1	NOT	XOR	XOR	XOR	NOT	XOR

Section C2

Control Signals			Functions for					
cs2	**cs1**	**cs0**	**Group 1**	**Group 2**	**Group 3**	**Group 4**	**Group 5**	**Group 6**
0	1	0	Transfer A	Sub	Increment A	Sub	Transfer A	Add
0	1	1	Sub with borrow	Sub with borrow	Add with carry	Add	Add with carry	Increment A
1	1	0	Decrement A	Add with carry	Transfer A	Increment A	Sub with borrow	Sub with borrow
1	1	1	Sub	Increment A	Add	Sub with borrow	Decrement A	Transfer A
X	0	0	XOR	NOT	NOT	AND	AND	OR
X	0	1	NOT	AND	XOR	OR	OR	XOR